

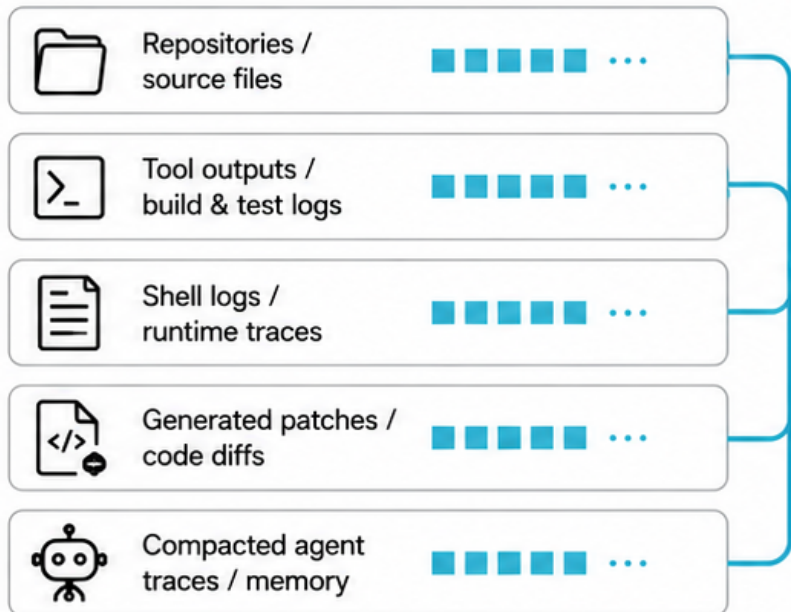
GLM-5.2: The 1M-Token Engineering Constraint

Figure 1 of 10

Workload Expansion

Token trajectory (single sequence)

≈200K tokens 1,048,576 tokens



Long-horizon coding and agentic execution

Scaling Pressure

DSA does not eliminate cache growth.

1 DSA indexer compute

$$Q_I K_I^T + \text{TopK}$$

repeated across layers

BOTTLENECK

2 KV / cache capacity

$$M_{\text{cache}} \propto BTL$$

Grows linearly with sequence length T across layers L and batch B

BOTTLENECK

3 Runtime overhead

- Long-context kernels
- Cache transfer (HBM ↔ HBM / CPU)
- CPU cache management
- Request scheduling
- GPU execution bubbles

BOTTLENECK

Reduced FLOPs do not imply proportional cache reduction.

GLM-5.2 Intervention Stack

1 IndexShare

reuse one index result across a four-layer region

Queries (Q_I)

DSA index (TopK result)

Layer ℓ

Layer $\ell + 1$

Layer $\ell + 2$

Layer $\ell + 3$

DERIVED

2 MTP IndexShare + KVShare

reuse first-step retrieval and target-model state across speculative steps

Step 1 retrieval

Target-model state (KV)

Speculative steps

2

3

...

n

DERIVED

3 Serving-engine optimisation

LayerSplit-derived memory management, cache-transfer coordination and CPU scheduling

Memory planning

Cache-transfer coordination

CPU scheduling

Kernel pipelining

DERIVED

Reported per-token FLOP reduction at 1M context = 2.9×

Reported model-level result; not equivalent to the raw sparse-attention ratio.



VERIFIED

Token / hidden-state flow

Attention computation

Residual addition

Reused cross-layer / cross-step state

Routing / optimisation control

Accepted / verified outputs

Implementation detail (not publicly disclosed)

Notation
(persistent)

B =batch size, T =sequence length, d =6144, H =64,
 $d_q=d_k=256=192+64$, $d_v=256$, $r_q=2048$, $r_{kv}=512$,
 $k_{DSA}=2048$, $E=256$, $K_{expert}=8$, $L=78$

Evidence badges

VERIFIED

Directly reported
in GLM-5.2 paper

DERIVED

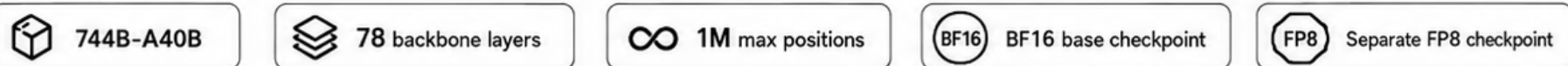
Logically derived
from reported facts

UNDISCLOSED

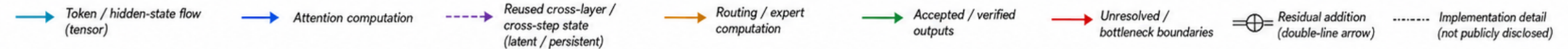
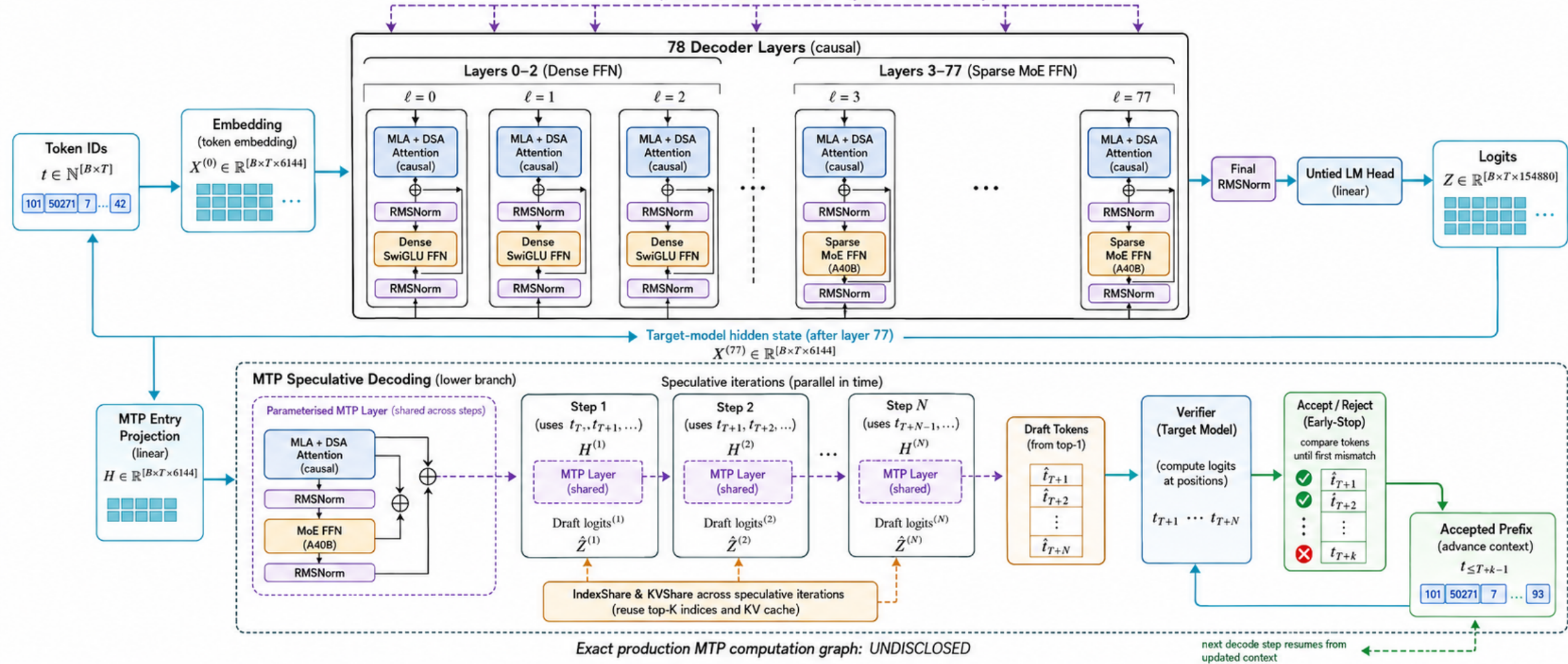
Implementation detail
not publicly disclosed

GLM-5.2 End-to-End Model Topology

Figure 2 of 10



$I^{(\ell)} \in \mathbb{N}^{[B \times T \times K]}$: TopK attention indices reused by shared-indexer layers



GLM-5.2 Input Representation and Position Encoding

Figure 3 of 10

A

Tokenizer boundary

Text / chat input

User: What is the capital of France?

Assistant:

Tokenizer



Token IDs $[B, T]$

154001 320 2674 13 483 ... 154827
0 1 2 3 4 ... $T-1$

UNDISCLOSED

Tokenizer algorithm

Pre-tokenisation regex

Merge vocabulary

Normalisation

• Vocabulary size $|V| = 154,880$

• Maximum position count = 1,048,576

• PAD ID = 154820

• EOS IDs = 154820, 154827, 154829

B

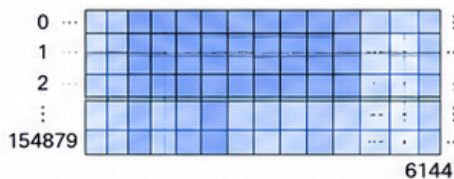
Embedding lookup

Token IDs $[B, T]$

154001 320 2674 ... 154827
0 1 2 ... $T-1$

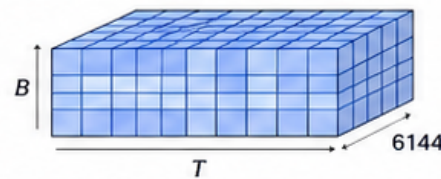
Input embedding matrix

$$E \in \mathbb{R}^{[154880 \times 6144]}$$



Looked-up hidden states

$$X^{(0)} = E[t] \in \mathbb{R}^{[B \times T \times 6144]}$$



Output projection (untied)

$$W_{LM} \in \mathbb{R}^{[6144 \times 154880]}$$



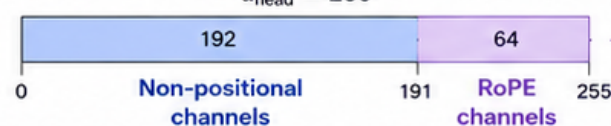
Input embedding and output projection are untied.

C

Model-attention positional split (per head)

One attention head (model space)

$d_{\text{head}} = 256$



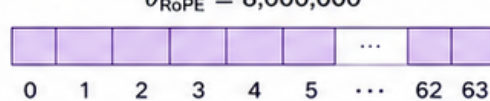
Non-positional channels: passed through unchanged

RoPE channels (64 dims): rotated with interleaved RoPE

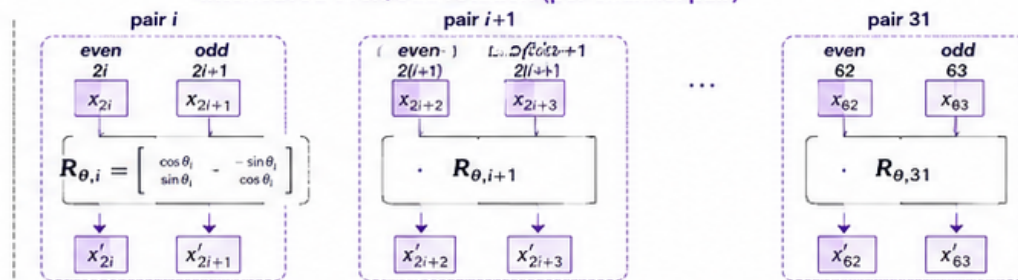
Only the 64-dim RoPE partition is rotated

RoPE channels (64 dims)

$\theta_{\text{RoPE}} = 8,000,000$



Interleaved even/odd rotation (per channel pair)



D

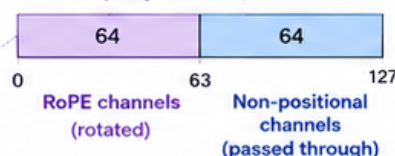
DSA indexer representation (uses interleaved RoPE)

DSA queries

$$Q_I \in \mathbb{R}^{[B \times T \times 32 \times 128]}$$

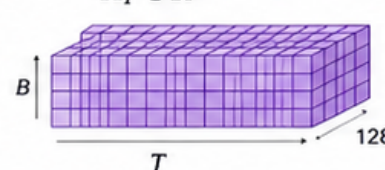


Per query vector (128 dims)

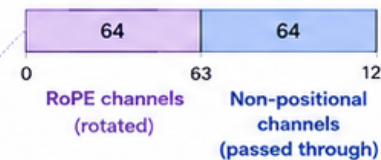


DSA keys (index vectors)

$$K_I \in \mathbb{R}^{[B \times T \times 128]}$$



Per key vector (128 dims)



Indexer configuration

Uses interleaved RoPE (same scheme as Panel C)

$\theta_{\text{RoPE}} = 8,000,000$

RoPE applied to the first 64 channels (interleaved even/odd), remaining 64 channels are non-positional.

→ Tokens / input flow



Hidden-state / model channels



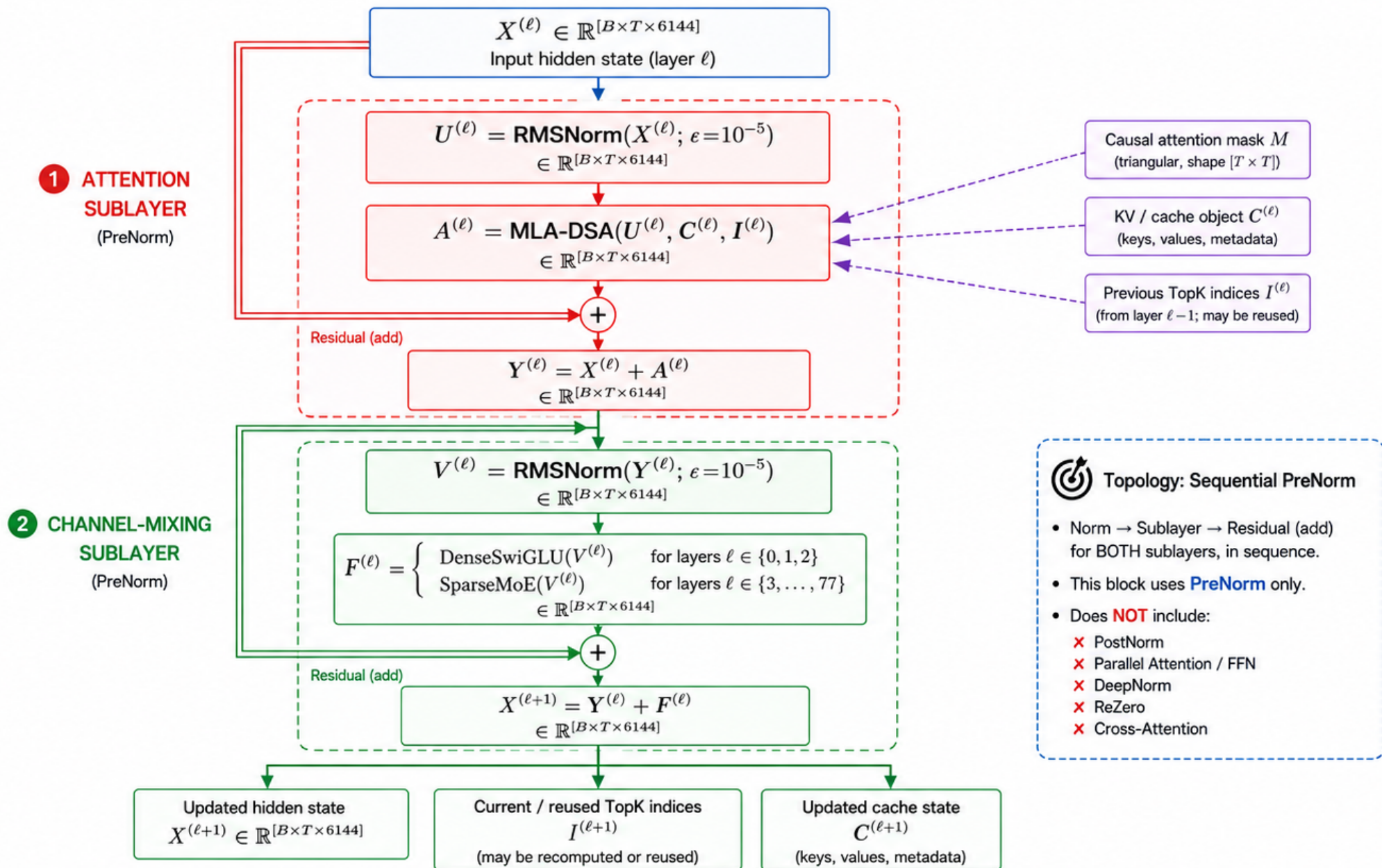
Positional / indexer channels



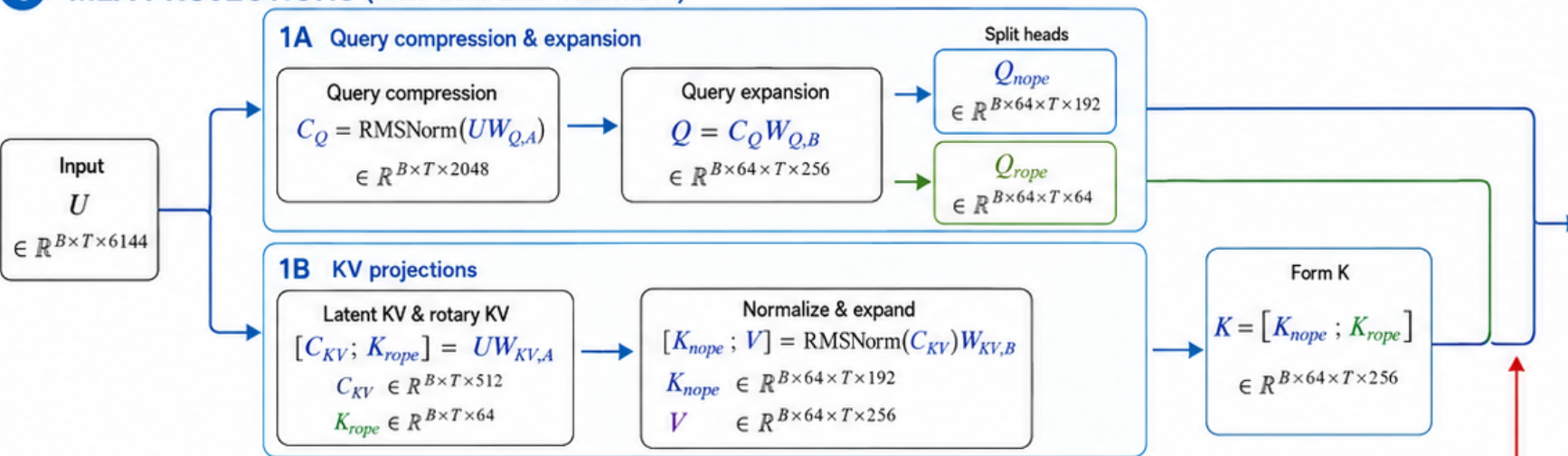
Undisclosed implementation details

GLM-5.2 Decoder Block: Exact Residual and Normalisation Order

Figure 4 of 10



1 MLA PROJECTIONS (Multi-head Latent Attention)



3 Converge: Sparse MLA Attention

For each head h and query position i :

Sparse attention over selected positions

Attend only to key/value positions in I_i

$$O_{i,h} = \sum_{j \in I_i} \text{softmax}\left(\frac{q_{i,h}^T k_{j,h}}{\sqrt{d_h}}\right) v_{j,h}$$

$$q_{i,h} \in \mathbb{R}^{d_h}, k_{j,h} \in \mathbb{R}^{d_h}, v_{j,h} \in \mathbb{R}^{d_v}$$

$$d_h = 256, d_v = 256 \text{ (per head)}$$

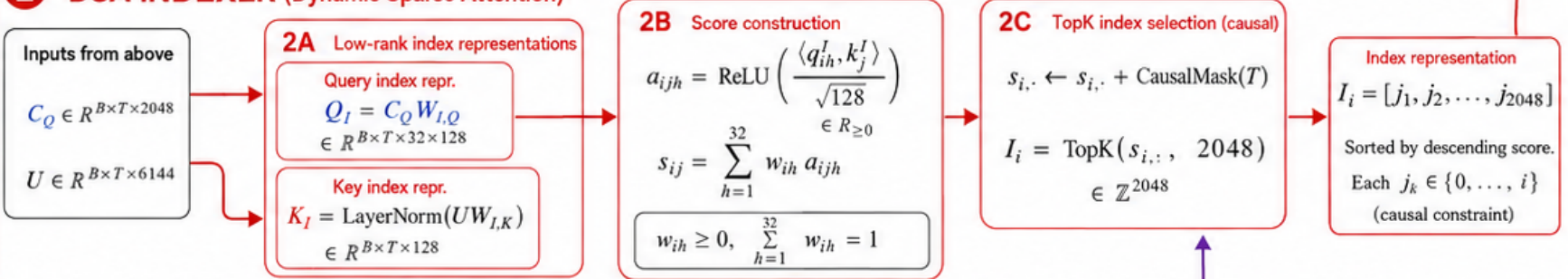
Output

$$O \in \mathbb{R}^{B \times 64 \times T \times 256}$$

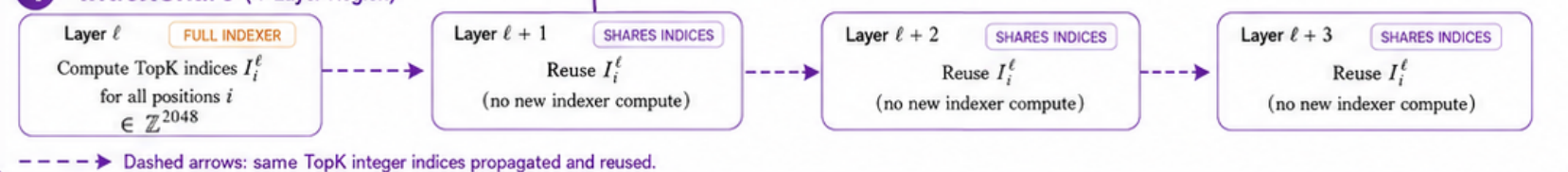
Dimensional legend

B batch size
 T sequence length
 $H = 64$ attention heads
 $d_h^{nope} = 192$
 $d_h^{rope} = 64$
 $d_h = 256$
 $d_v = 256$

2 DSA INDEXER (Dynamic Sparse Attention)



4 IndexShare (4-Layer Region)



Backend boundary

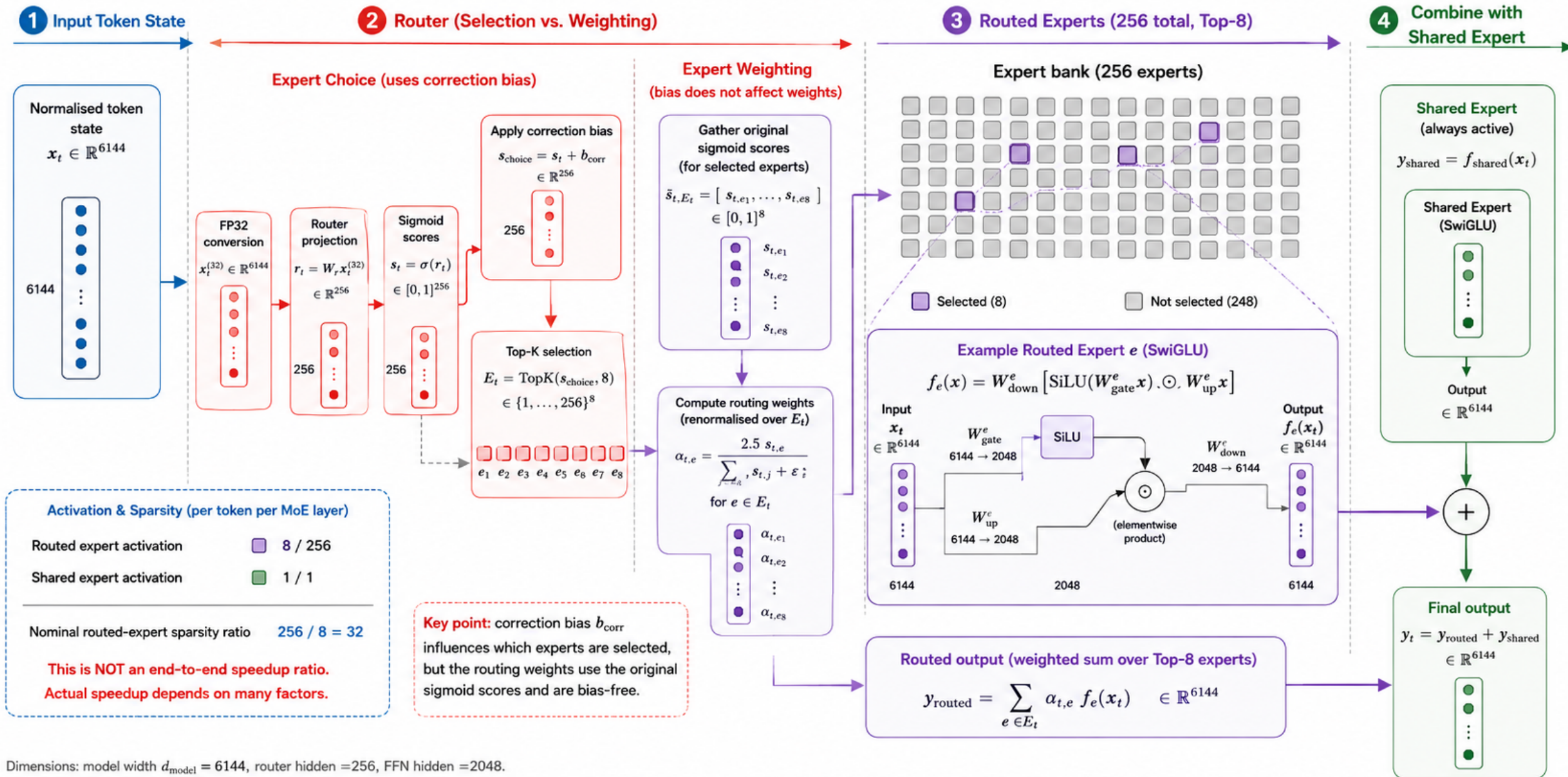
The TopK integer indices I_i are the primary interface.

- Eager (reference) or SDPA path**
convert I_i to an additive sparse mask M_i for standard attention kernels.
 $M_i[j] = 0$ if $j \in I_i$, $M_i[j] = -\infty$ otherwise.
- Specialised sparse kernel path**
consume I_i directly to gather K/V and compute attention without materialising a dense mask.

Exact production kernel implementation:

UNDISCLOSED

GLM-5.2 Sparse-MoE Token Routing



Dimensions: model width $d_{\text{model}} = 6144$, router hidden = 256, FFN hidden = 2048.

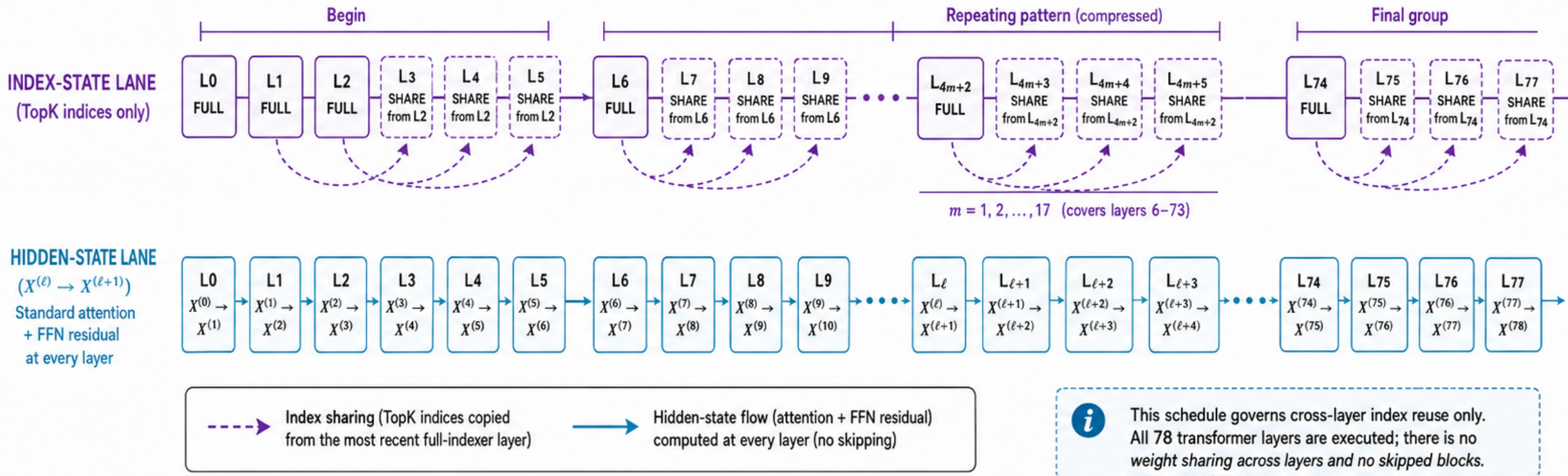
$W_r \in \mathbb{R}^{256 \times 6144}$, $b_{\text{corr}} \in \mathbb{R}^{256}$, TopK = 8, scaling factor = 2.5, ε is a small constant for numerical stability.

GLM-5.2 Cross-Layer IndexShare Schedule

Figure 7 of 10



Depth timeline across all 78 layers (0–77). Indexers are computed only on scheduled full-indexer layers. Subsequent three layers reuse (share) those indices for retrieval.



DERIVED FROM CONFIGURATION

21

Full-indexer layers
(independent index compute)

+

57

Shared-indexer layers
(reuse indices from previous
full-indexer layer)

=

$1 - \frac{21}{78} = 73.08\%$
of backbone layers do not independently
execute a full indexer



This percentage is not a measured latency reduction.

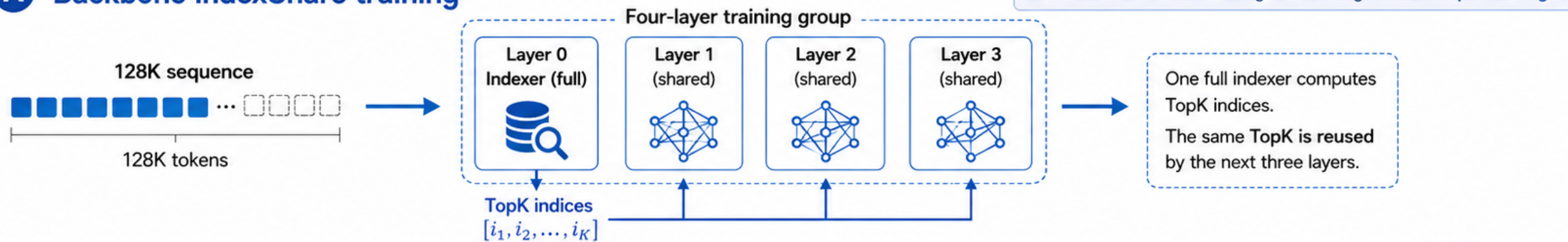
It reflects the portion of layers that reuse indices and therefore do not run an independent indexer.

-----> Index sharing (TopK indices copied from the most recent full-indexer layer)

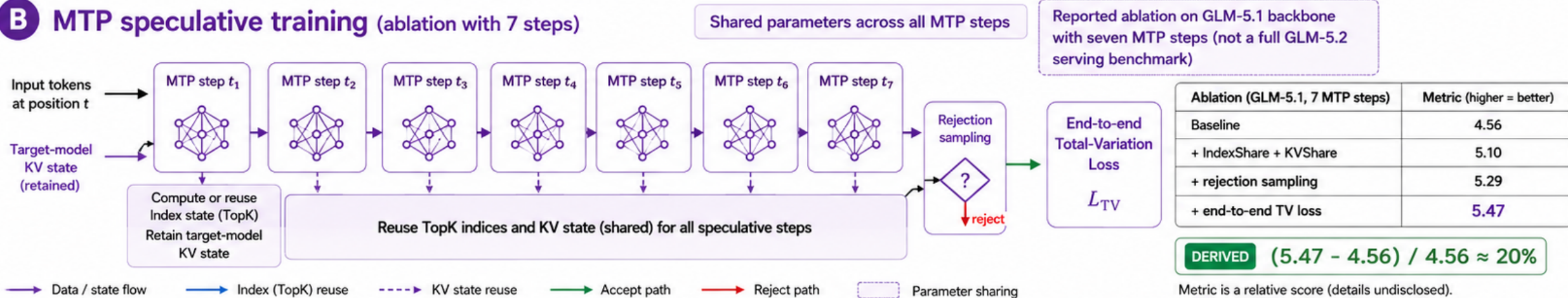
GLM-5.2 Training and Speculative-Decoding Optimisation

Figure 8 of 10

A Backbone IndexShare training



B MTP speculative training (ablation with 7 steps)



C Precision matrix

Component	Precision / Format	Notes	Component	Precision / Format	Notes
Released base checkpoint	BF16	Default weight checkpoint format for released model.	Indexer weight projection	Retained in FP32	Indexer projection matrices are kept in FP32.
Separate checkpoint	FP8	Optional separate FP8 weight checkpoint for inference.	Cache precision	UNDISCLOSED	Key / value cache precision not publicly disclosed.
MoE router logits	FP32	Router logits are computed and stored in FP32.	Optimizer states	UNDISCLOSED	Optimizer state precision not publicly disclosed.
Correction bias	FP32	Bias corrections (e.g., residual calibration) in FP32.	FP8 scale format / grouping	UNDISCLOSED	FP8 scale format and grouping not publicly disclosed.

UNDISCLOSED = implementation detail not publicly disclosed.

GLM-5.2 1M-Context Inference Dataflow

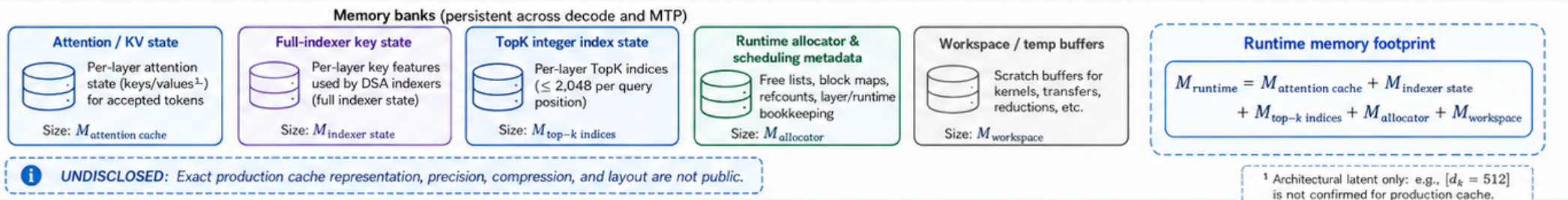
Figure 9 of 10

1 Request and prefill

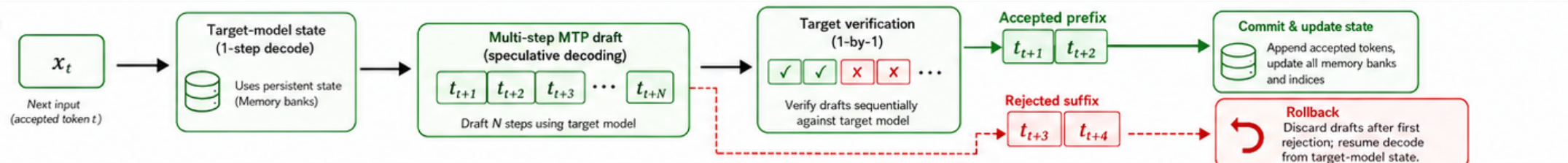


2 Persistent state

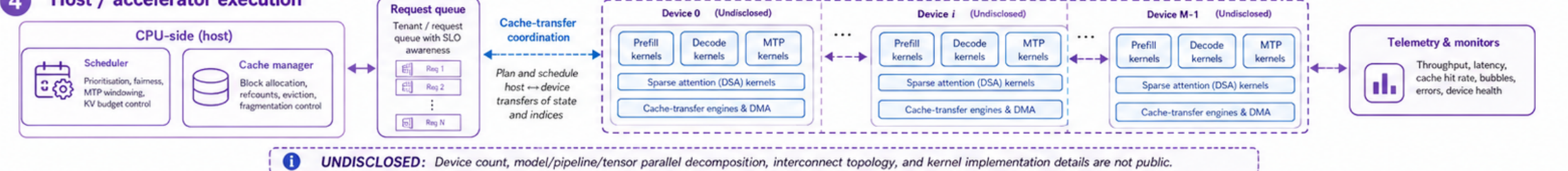
Per-tenant (or request) state persisted across decoding steps and MTP iterations.



3 Decode and MTP



4 Host / accelerator execution



Release-level optimisation themes



Finer-grained memory management & parallelisation derived from LayerSplit



Cache-transfer & kernel coordination across stages/devices



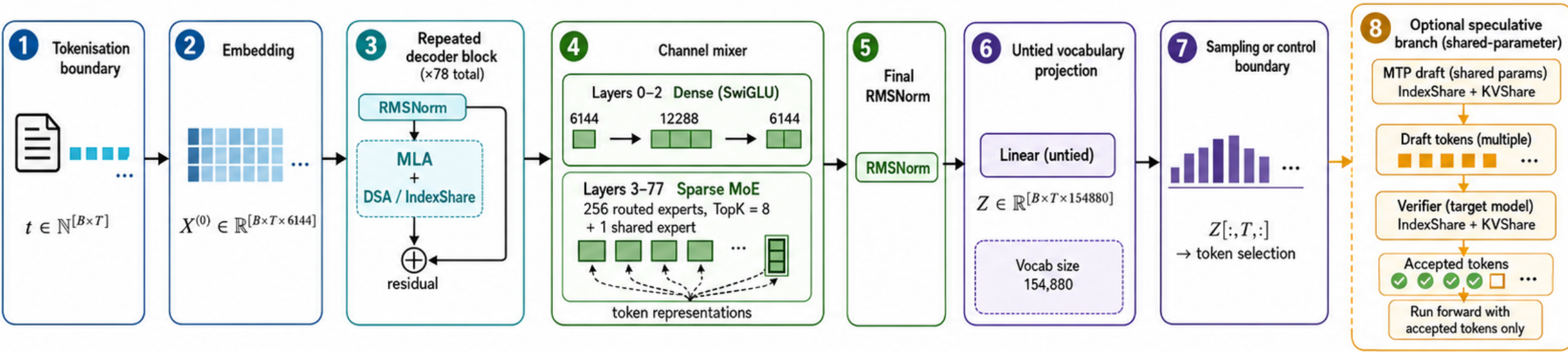
CPU cache management & request scheduling tuned for long-context workloads



Reduction of GPU pipeline bubbles via balanced execution

GLM-5.2 Integrated Forward Pass and Engineering Trade-Offs

Figure 10 of 10



744B total / ~40B active

78 backbone layers

Context limit 1,048,576

64 MLA heads

DSA TopK 2,048

256 routed experts / Top-8

DERIVED

 21 full + 57 shared indexer layers

BF16

FP8 checkpoints released separately

Mechanism	Direct benefit	Remaining cost or constraint
MLA	Latent Q/KV projection structure reduces per-head KV size and improves bandwidth.	Runtime cache layout, memory format, and fused-kernel efficiency remain backend-dependent.
DSA	Attends over selected positions, scaling compute with TopK rather than full context.	Index construction and validation still incur long-context cost and latency.
IndexShare	Removes repeated indexer work in shared layers, amortising index computation.	Reused indices constrain later layers to the same selected set of positions.
Sparse MoE	Activates 8 of 256 routed experts (Top-8) per token, reducing FFN compute.	Routing, shared expert compute, and distributed communication remain.
MTP	Generates multiple candidate tokens and verifies them in parallel.	Acceptance rate is workload- and implementation-dependent.
1M context	Enables longer, uninterrupted trajectories and fewer resets.	Cache capacity, memory bandwidth, and orchestration complexity dominate.
Open weights / config	Architecture, hyperparameters, and data processing are inspectable.	Production training and serving stack (infra, kernels, system optimizations) remains partially undisclosed.

Scientific conclusion: GLM-5.2 primarily restructures where expensive retrieval indices and speculative state are recomputed. It reduces repeated computation, but *does not remove long-context memory and systems constraints*.